AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	MD facility MD_way_178331621 -- station KJYO, America/New_York
Building	OSM 178331621
	OSM way 178331621
Facade gap	not applicable -- one building, no second facade
Weather record	43,249 real hourly records from KJYO
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-03-25 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 23 hours with 2 mode changes. 7 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	10 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
	7 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	8.000	18.0	9.000	3.757
01	FREE	yes	-	8.000	18.0	9.000	3.105
02	FREE	yes	-	9.000	18.0	9.000	3.453
03	FREE	yes	-	9.000	18.0	9.000	2.956
04	FREE	yes	-	9.000	18.0	10.000	2.620

05	FREE	yes	-	9.000	18.0	9.000	2.241
06	FREE	yes	-	9.000	18.0	10.000	1.824
07	FREE	yes	-	11.000	18.0	10.000	2.626
08	FREE	yes	-	13.000	18.0	10.000	2.642
09	FREE	yes	-	16.000	18.0	10.000	3.702
10	mechanical	no	dry-bulb	20.000	18.0	10.000	5.013
11	mechanical	no	dry-bulb	21.000	18.0	11.000	5.612
12	mechanical	no	dry-bulb	23.000	18.0	11.000	5.658
13	mechanical	no	dry-bulb	23.000	18.0	11.000	5.729
14	mechanical	no	dry-bulb	21.000	18.0	12.000	4.865
15	mechanical	no	dry-bulb	19.000	18.0	13.000	4.426
16	mechanical	yes	switch budget	16.000	18.0	14.000	3.163
17	mechanical	yes	switch budget	15.000	18.0	14.000	3.002
18	mechanical	yes	switch budget	13.000	18.0	14.000	2.792
19	mechanical	yes	switch budget	12.000	18.0	13.000	3.554
20	mechanical	yes	switch budget	11.000	18.0	13.000	3.004
21	mechanical	yes	switch budget	12.000	18.0	12.000	3.892
23	mechanical	yes	switch budget	13.000	18.0	13.000	4.198

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.000 C, which is 10.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.757 C, and the margin is measured, not chosen: 3.757 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.000 C, which is 10.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.105 C, and the margin is measured, not chosen: 3.105 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.000 C, which is 9.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.453 C, and the margin is measured, not chosen: 3.453 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.000 C, which is 9.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.956 C, and the margin is measured, not chosen: 2.956 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.000 C, which is 9.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.620 C, and the margin is measured, not chosen: 2.620 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.000 C, which is 9.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.241 C, and the margin is measured, not chosen: 2.241 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.000 C, which is 9.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.824 C, and the margin is measured, not chosen: 1.824 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.000 C, which is 7.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.626 C, and the margin is measured, not chosen: 2.626 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.000 C, which is 5.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.642 C, and the margin is measured, not chosen: 2.642 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.000 C, which is 2.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.702 C, and the margin is measured, not chosen: 3.702 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.000 C against a 18.0 C limit, so it fails by 2.000 C. It would take a limit 2.000 C higher, or a bound 2.000 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.000 C against a 18.0 C limit, so it fails by 3.000 C. It would take a limit 3.000 C higher, or a bound 3.000 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.000 C against a 18.0 C limit, so it fails by 5.000 C. It would take a limit 5.000 C higher, or a bound 5.000 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.000 C against a 18.0 C limit, so it fails by 5.000 C. It would take a limit 5.000 C higher, or a bound 5.000 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.000 C against a 18.0 C limit, so it fails by 3.000 C. It would take a limit 3.000 C higher, or a bound 3.000 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 19.000 C against a 18.0 C limit, so it fails by 1.000 C. It would take a limit 1.000 C higher, or a bound 1.000 C tighter, to change this hour.

16:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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